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Unified Sync-Async Coupled Dynamical Systems Approach with Obstacle-Aware Modulation for Stable Bimanual Handover

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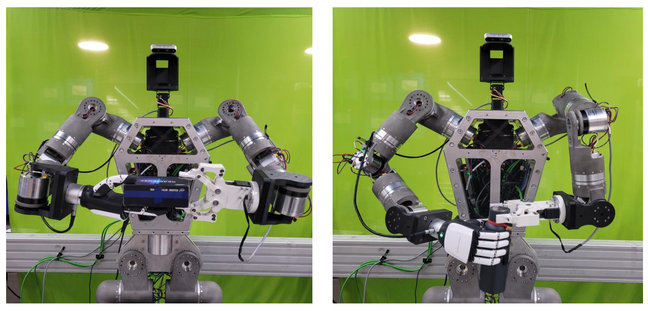

Fig. 1: Demonstration of the proposed framework for handover on an upper torso dual-arm platform.

Introduction

Bimanual handover is a **common primitive** across industrial assembly, warehouse transfer, and humanoid manipulation. In this work we address the problem how can two robotic manipulators perform an **object handover in real-time** such that they (i) **converge to an identical handover pose at precisely the same instant**, and (ii) **maintain this temporal-spatial synchrony and coordination**, even under disturbances.

Our Approach

We propose a unified dynamical-systems (DS) controller that replaces **segmented pipeline stages with a single formulation**. The method combines asynchronous reaching into coupled synchrony via a **consensus clock** shared by translational and quaternion-log orientation DS. An **obstacle-aware modulation** layer deflects velocities around obstacles **without replanning**, while preserving stability and the global attractor.

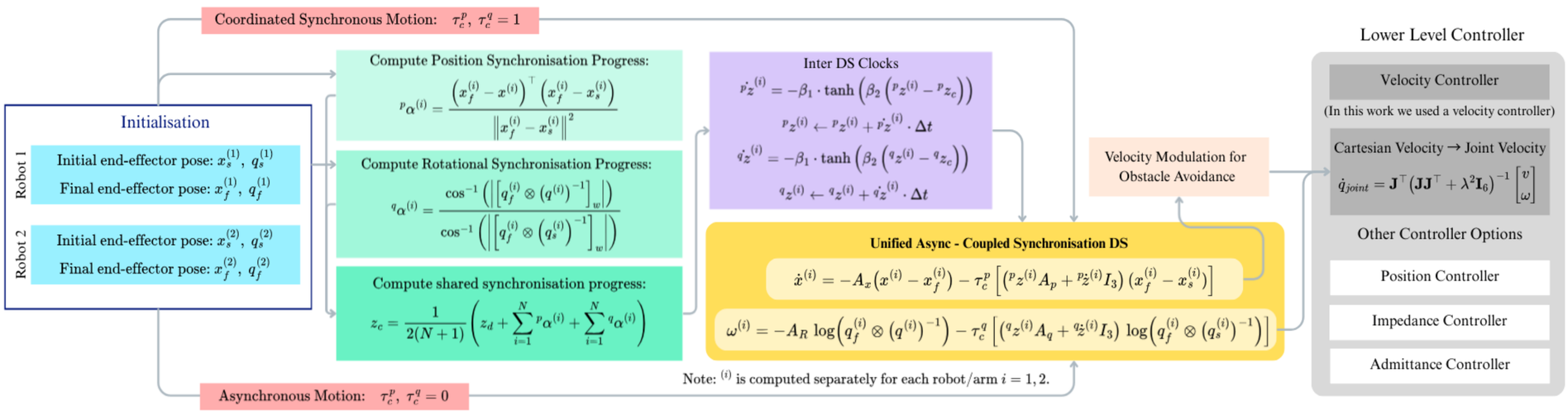

Fig. 2: Overview of the proposed, consensus driven dynamical systems framework

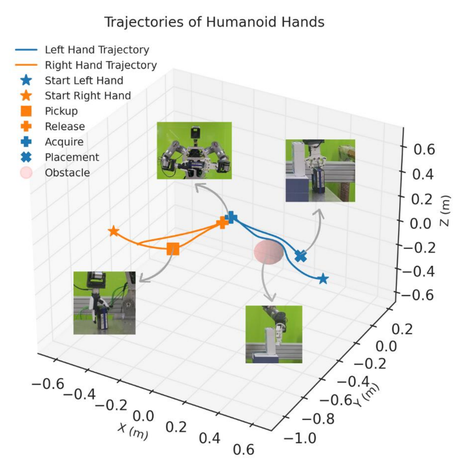

Fig. 3: Hand-over end effector trajectories.

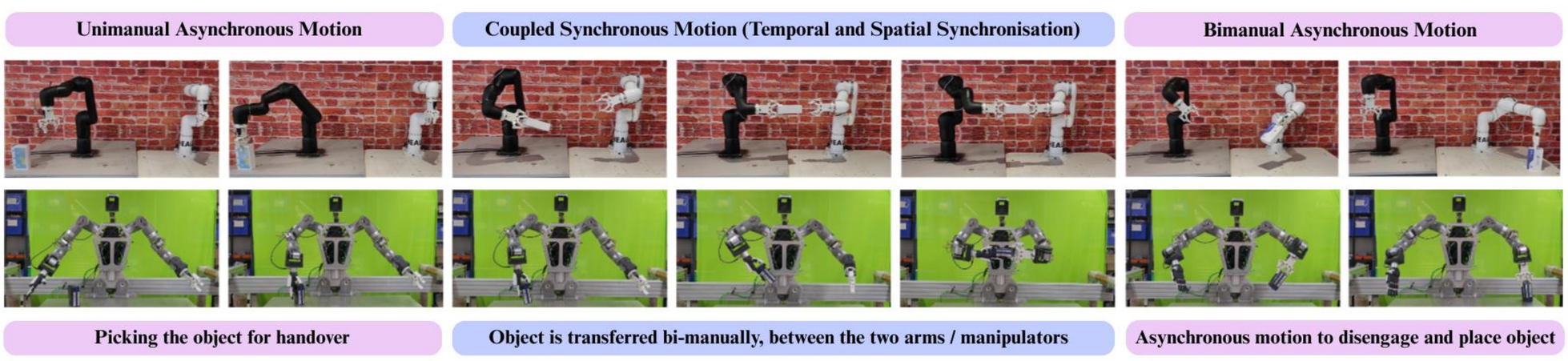

Fig. 4: Snapshots of the proposed handover framework across different phases.

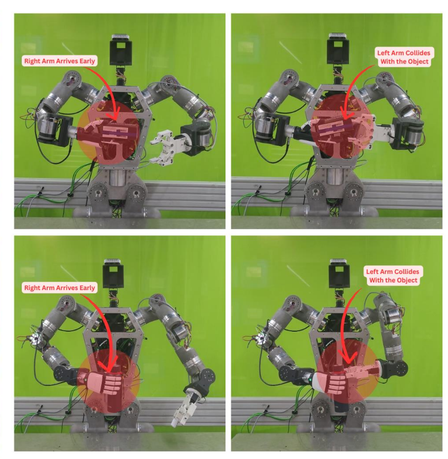

Fig. 5: Failure cases under ablation study.

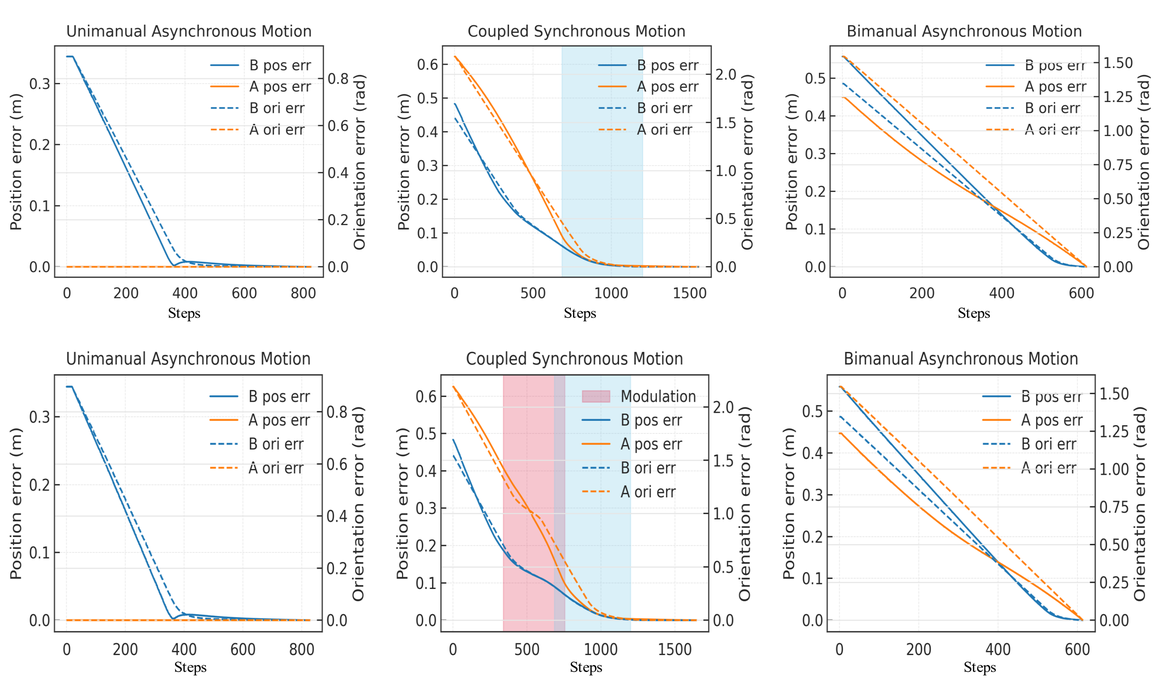

Fig. 6: Error convergence plots of the proposed framework, across handover of objects. Blue bands mark the synchronous handover window; red highlights indicating obstacle-induced velocity modulation.

Results – Experimental Implementation on Two Platforms

- Position and orientation errors **co-decay in the same window**; no hovering.
- Synchrony holds even with obstacle-induced detours.
- Ablation (no coupling): early arrival, desynchrony, collisions.
- **Success:** with coupling **10/10**; without **1/10** (box: **5/5 vs 0/5**, bottle: **5/5 vs 1/5**).

Future Work

- **Phase-sequenced coupling:** position → orientation (e.g., tilt–pour) using phase-conditioned gates.
- Generalize to more bimanual tasks; **DS low-level + VLM/LLM high-level**.